MODULAR CONTRACTS

Jason Roderick Donaldson   Giorgia Piacentino   Zichen Zhao

FACTS

Current theory: Contract as incentive/sharing rule: outcome $\mapsto$ transfers

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Real contracts

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Real contracts: Written in language and divided into modular clauses

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Evaluated clause by clause

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- <u>Complexity risk</u>

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- <u>Incompleteness</u>: Clauses omit relevant contingencies
- <u>Conservatism</u>: High burden of truth to verify clause

QUESTION

How can model of modular contracts explain such contractual aberrations?

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Model of modular contracts

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Model of modular contracts: Modules = clauses

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Clauses collections of atomic premises

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- Clauses collections of atomic premises

- Premises evaluated (1 or 0) based on threshold of observed data

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Key assumption

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Marginal unchanged $\implies$ evaluate $\mathbf{a}_j, \mathbf{a}_k$ correctly

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Marginal unchanged $\implies$ evaluate $\mathbf{a}_j, \mathbf{a}_k$ correctly, not joint $\mathbf{a}_j \wedge \mathbf{a}_k$

RESULTS

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Narrow clauses

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Extension: High thresholds $\Longrightarrow$ conservatism

MODEL

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(“Atomic”) Premises

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<u>("Atomic") Premises</u>: $\mathcal{A} = \{\mathbf{a}_j\}_j$, $\mathbf{a}_j = (a_j^1 ... a_j^I) \in \{0,1\}^I$ is vector of <u>data</u>

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<u>Objective</u>

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<u>Objective</u>: Minimize $C = c_{\mathrm{I}} \mathsf{Pr}[[\![\Phi]\!] > [\![\phi_{\mathcal{T}}]\!]] + c_{\mathrm{II}} \mathsf{Pr}\,[[\![\Phi]\!] < [\![\phi_{\mathcal{T}}]\!]]$

MODEL: EXAMPLE: SHOE DELIVERY

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Contract Φ: E.g. "right shoe" $\wedge$ "left shoe"; e.g. "pairs"

MODEL: EXAMPLE: SHOE DELIVERY

("Atomic") Premises: $\mathbf{a}_\ell$: left shoe $\mathbf{a}_r$: right; Data: size $i \in \{1, 2, 3\}$

Target policy: $\phi_{\mathcal{T}} = \mathbf{a}_\ell \wedge \mathbf{a}_r$ ("[deliver] pairs": $a_\ell^i \wedge a_r^i$)

Observations = shuffled data: $\hat{\mathbf{a}}_j$ sizes shuffled w.p. e

Clause: "right shoe," "left shoe," "pairs"

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Objective: $C = c_{\mathrm{I}}\mathsf{Pr}[\text{accept} < \text{two pairs}] + c_{\mathrm{II}}\mathsf{Pr}[\text{reject} > \text{two pairs}]$

MODEL: EXAMPLE: COVENANTS

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Premises

MODEL: EXAMPLE: COVENANTS

Premises: $\mathbf{a}_{\mathrm{IC}}$: "interest coverage", $\mathbf{a}_{\$}$: "cash"; Data: date-$i$ statements

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Target policy

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Contract Φ

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<u>Shuffle risk</u> states `SR`: Data realizations s.t. $\mathsf{Pr}\Big[[\![\Phi^{\mathcal{A}}]\!] \neq [\![\Phi^{\mathcal{T}}]\!] \;\Big|\; \{\mathbf{a}_j\}_j\Big] > 0$

EXAMPLE

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Two premises with three data each and $\tau = 2$:

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i
1
2
3

EXAMPLE

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Loophole: Evaluation & conjunction don't commute: $[\![\mathbf{a}_j]\!] \wedge [\![\mathbf{a}_k]\!] \neq [\![\mathbf{a}_j \wedge \mathbf{a}_k]\!]$

EXAMPLE: SHUFFLE RISK

Observe permuted data

i	$\mathbf{a}_j$	$\mathbf{a}_k$	$\mathbf{a}_j \wedge \mathbf{a}_k$
1	1	1	1
2	1	0	0
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# of 1s $\geq \tau$	**1**	**1**	**0**

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Observe permuted data E.g. $\hat{\mathbf{a}}_j = \mathbf{a}_j$ (no shuffle)

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Observe permuted data E.g. $\hat{\mathbf{a}}_j = \mathbf{a}_j$ (no shuffle) ; $\hat{\mathbf{a}}_k = 110$ (shuffle of $\mathbf{a}_k$)

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Joint changed: $101 \to 110 \Rightarrow$ target-based evaluation flips

RESULTS

LEMMA

LEMMA: LOOPHOLES $\subsetneq$ SHUFFLE RISK

$\texttt{LH} \subsetneq \texttt{SR}$

LEMMA: LOOPHOLES $\subsetneq$ SHUFFLE RISK: PROOF

$\texttt{LH} \Rightarrow \texttt{SR}$

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But shuffle could be $\hat{\mathbf{a}}_j = 110, \hat{\mathbf{a}}_k = 101$: $[\![\hat{\mathbf{a}}_j \wedge \hat{\mathbf{a}}_k]\!] = [\![100]\!] = 0$ (yes $\texttt{SR}$)

R1: LOOPHOLES

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With $e > 0$, if $\mathsf{Pr}[\mathtt{SR}]$ large enough relative to $\mathsf{Pr}[\mathtt{LH}]$, then $\Phi^{\mathcal{A}} \succ \Phi^{\mathcal{T}}$

R1: LOOPHOLES: INTUITION

Shuffled data ($e > 0$): Evaluation errors for all but not only loopholes

Accept loopholes to reduce shuffling error

LOOPHOLES IN PRACTICE: J. CREW

Distressed J. Crew wanted to free collateral

Credit agreement ostensibly prohibited moving it to unrestricted sub, but...

Clause $\mathbf{a}_j$: "Can move assets to restricted sub"

Clause $\mathbf{a}_k$: "Can move assets from restricted to unrestricted sub"

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Clause $\mathbf{a}_k$: "Can move assets from restricted to unrestricted sub"

$\Rightarrow \mathbf{a}_j \wedge \mathbf{a}_k$: "Can move assets to unrestricted sub [via unrestricted]"

I.e. loophole: $[\![\mathbf{a}_j]\!] \wedge [\![\mathbf{a}_k]\!] \neq [\![\mathbf{a}_j \wedge \mathbf{a}_k]\!]$

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Katz 10: Many loopholes openly exploited after much litigation

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Doctrinal Paradox (DP): Vote on premises $\neq$ on targets (List–Pettit 02)

LOOPHOLES IN THE LITERATURE

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Show why loopholes persist $e > 0$

R2: COMPLEXITY RISK

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If $c_{\mathrm{I}}\mathsf{Pr}[\mathtt{LH}]$ sufficiently high, $\Phi^{\mathcal{T}} \succ \Phi^{\mathcal{A}}$

R2: COMPLEXITY RISK: INTUITION

If loopholes likely or costly use complex target-based contract (long clauses)

Cost: With interacting premises, shuffle risk $\Rightarrow$ likely evaluated with error

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Empirically: Complexity leads to errors in taxes and regulator compliance
OECD, Benzarti 20, Zwick 21, Behn et al 22...

R3: REDUNDANCY

DEFINITION: REDUNDANCY

$\Phi^{\mathcal{R}}$ called redundant if $j \in \mathcal{M}_m \cap \mathcal{M}_{m'}$ for some clauses $j \in \mathcal{A},\ m \neq m'$

R3: REDUNDANCY

If $e > 0$ not too high and $c_{\mathrm{I}}\mathsf{Pr}[\mathtt{LH}]$ high enough, both $\Phi^{\mathcal{R}} \succ \Phi^{\mathcal{T}}$ & $\Phi^{\mathcal{R}} \succ \Phi^{\mathcal{A}}$

R3: REDUNDANCY: INTUITION

Benefit w.r.t. target based: Fewer false positives

Repeated clause $\Rightarrow$ require multiple shuffling errors

Benefit w.r.t. premise-based: Fewer loopholes

Keeps joint information

R3: REDUNDANCY: INTUITION

Benefit w.r.t. target based: Fewer false positives

Repeated clause $\Rightarrow$ require multiple shuffling errors

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Keeps joint information

Cost: False negatives: Shuffles in any clause can cause rejection

R4: INCOMPLETENESS

DEFINITION: INCOMPLETENESS

$\Phi^{\mathcal{I}}$ called incomplete w.r.t. $\mathcal{T}$ if $j \notin \bigcup \mathcal{M}_m$ for some $j \in \mathcal{T}$

R4: INCOMPLETENESS

If e and c_{II} high and $\mathsf{Pr}\left[\llbracket\Phi^{\mathcal{I}}\rrbracket > \llbracket\phi^{\mathcal{T}}\rrbracket\right]$ low, then $\Phi^{\mathcal{I}} \succ \Phi^{\mathcal{T}}$

R4: INCOMPLETENESS: INTUITION

Long clauses induce shuffling risk $\Rightarrow$ omit some premises to avoid it

Akin to mismeasurement in Holmström–Milgrom:

Decrease power on distortionary tasks

R5: CONSERVATISM

EXTENSION

Clause-dependent thresholds $\tau_{\mathcal{M}}$

DEFINITION: CONSERVATISM

Let $\tau_{\mathcal{T}}$ be threshold of target

$\Phi^{\mathcal{C}}$ is <u>conservative</u> if $\tau_{\mathcal{M}_m} > \tau_{\mathcal{T}}$ for every m

R5: CONSERVATISM

Define $\texttt{ACC}_\tau(\Phi) := \big\{\{\mathbf{a}_j\}_j \ : [\![\Phi]\!] = 1\big\}$

If $\mathsf{Pr}\big[[\![\phi_{\mathcal{T}}]\!] = 1 \mid \texttt{ACC}_\tau(\Phi^{\mathcal{A}}) - \texttt{ACC}_{\tau+1}(\Phi^{\mathcal{A}})\big] < \dfrac{c_{\mathrm{I}}}{c_{\mathrm{I}} + c_{\mathrm{II}}}$, then $\Phi^{\mathcal{C}} \succ \Phi^{\mathcal{A}}$

R5: CONSERVATISM: INTUITION

Premise-based evaluations gives too many false positives

Conservatism decreases them via increased threshold

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Premise-based evaluations gives too many false positives

Conservatism decreases them via increased threshold

Feature: Relies on only marginals $\Rightarrow$ not subject to shuffle risk

R5: CONSERVATISM: APPLICATIONS

Law: "beyond reasonable doubt," "reasonable suspicion," "probable cause"
In re Winship, 397 U.S. 358 (1970), Terry v. Ohio, 392 U.S. 1 (1968)

Accounting & auditing: High standard for good news Basu 97; Watts 03

E.g. losses recognized faster than gains

CONCLUSION

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Model of modular contracts in formal language s.t. shuffled data

- Loopholes
- Complexity risk
- Landmines
- Incompleteness
- Conservatism